

A Two Experiment Treatment Comparison Study: Teaching Social Skills to Children with Autism Spectrum Disorder

Debra Berry Malmberg · Marjorie H. Charlop ·
Sara J. Gershfeld

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Abstract The present study is a series of two experiments of treatment comparisons addressing the acquisition of social skills for children with Autism Spectrum Disorder (ASD). In Experiment 1, a multi-element design was used to assess the efficacy of video modeling as compared with Social Stories™ to teach various social skills to the children with ASD. Results from video modeling yielded positive and quick acquisition, consistent with previous literature. However, treatment with Social Stories did not. Upon close inspection of the Social Stories literature, it became apparent that most effective social stories interventions were treatment packages, which include well-tested behavioral procedures such as prompting. Therefore, Experiment 2 was designed to compare Social Stories with the components most often used in social stories treatment packages, such as prompting, parceled out, to assess variables functional for behavior change. A multiple baseline design across children was used. Results of Exp. 2 suggested that prompting was effective for the children in the present study. Results of both experiments are discussed in terms of relating these results to previous literature, and ultimately to recommending treatment for social skills for children with ASD.

Keywords Autism Spectrum Disorder · Treatment comparison · Social skills · Social behavior · Video modeling · Social stories · Prompting

D. B. Malmberg (✉)
Department of Psychology, California State University, Northridge, 18111 Nordhoff Street, Northridge,
CA 91330-8255, USA
e-mail: dmalmberg@csun.edu

M. H. Charlop
Department of Psychology, Claremont McKenna College, Claremont, CA 91711, USA

S. J. Gershfeld
Department of Psychology, Scripps College, Claremont, CA 91711, USA

As children with Autism Spectrum Disorder (ASD) continue to benefit from Applied Behavior Analysis and integration with neurotypical children becomes more prevalent, the need for evidence-based social skills treatment becomes more urgent (Tomaino, Miltenberger, and Charlop 2014). One treatment that has been shown in the literature to be effective with social behaviors is video modeling (VM). There are many possible reasons why VM is an appealing and effective treatment approach to use with children with ASD. First, VM minimizes attention and language behaviors, requiring the child only to look at a small spatial area (i.e., a television, computer, or a handheld tablet) and to hear only the minimum necessary language. Second, many children with ASD frequently and spontaneously imitate videos (Charlop-Christy and Danesvhar 2003; Shane and Albert 2008), a tendency upon which video modeling interventions capitalize. Third, creating footage of models is relatively simple and efficient. For instance, Alberto and colleagues (2005) found that developing video materials is less time consuming than developing many photographic learning tools (Alberto, Cihak, & Gama). Also, use of a VM intervention is relatively economical in terms of resources and materials, especially when compared with interventions that use in vivo instruction (Gena, Couloura, and Kymissis 2005; Shipley-Benamou, Lutzker, and Taubman 2002). Finally, video footage can be used repeatedly with the same child and possibly for other children with similar deficits. Perhaps the most frequently discussed reason why video modeling is efficacious is because it is a visually-based learning tool which seems to be of preference as well as of benefit to learners with ASD (e.g., Ayres and Langone 2005; Charlop and Milstein 1989; Delano 2007).

Numerous studies have demonstrated VM as an effective intervention in teaching social skills such as conversational speech (Charlop and Milstein 1989; Charlop, Dennis, Carpenter, and Greenberg 2010; Taylor, Levin, and Jasper 1999), imaginative play (D'Ateno, Mangiapanello, and Taylor 2003; MacDonald, Clark, Garrigan, and Vangala 2005), and perspective-taking skills (Charlop-Christy and Danesvhar 2003; LeBlanc et al. 2003) to children with ASD. However, only one published study to date was a direct treatment comparison, which compared video modeling to in vivo (live) modeling (Charlop-Christy, Le, and Freeman 2000). The results from this study found that that video modeling led to faster acquisition than the in vivo condition. Additionally, greater generalization of the skills occurred with VM, and the cost-benefit analysis yielded VM more efficient than in vivo in terms of time, money, and personnel.

Additional treatment comparison studies are needed to identify the most efficient and effective interventions to teach social skills for children with ASD – especially those skills that will be necessary in natural environments such as home and school or with siblings and peers. Whereas VM is a likely choice of treatment, another oft-cited treatment used with children with ASD is Social Stories™ (cf. Gray 2004; Gray and Garand 1993). Social Stories are narratives written from a first-person perspective that describe the context, the appropriate responses in that situation, other people's perspectives on appropriate behavior, and the value of these behaviors (Gray 2004; Gray and Garand 1993). Children with ASD listen to the stories, or read the stories themselves, then answer brief questions about the story content before being given the opportunity to respond appropriately. Research has supported the use of Social Stories treatments in teaching social behaviors for children with ASD such as initiations, commenting, social engagement, and conversational skills (e.g., Crozier and Tincani 2007; Delano and

Snell 2006; Dodd, Hupp, Jewell, and Krohn 2008; Sansosti and Powell-Smith 2006; Scattone, Tingstrom, and Wilczynski 2006).

In the present study, Experiment 1 will expand upon previous VM literature by: 1) addressing social skills that have been less researched in the literature and 2) providing a direct treatment comparison with Social Stories. However, as Chan and O'Reilly (2008) as well as other researchers (Iskander and Rosales 2013; Reynhout and Carter 2011a; Test, Richter, Knight, and Spooner 2011), have pointed out, it is unclear what aspects of Social Stories, per se, have been functional in behavior change. Previous literature on Social Stories has frequently combined additional procedures such as prompting, modeling, and/or reinforcement. Thus, in Experiment 1, our Social Stories treatment comparison will consist of Social Stories solely rather than as part of a treatment package. Experiment 2 will further explore this question by providing another treatment comparison study in which Social Stories solely (Gray and Garand 1993; Gray 2004) will be compared with typical behavioral procedures that are often added as treatment package components with Social Stories (e.g., Kokina and Kern 2010; Reynhout and Carter 2009).

Experiment 1

As previously discussed, VM is an evidence-based treatment often used with children with ASD (Charlop-Christy et al. 2000). As a means to propel the literature on social skills, this body of work was extended to study additional target social skills. Additionally, a treatment comparison with an oft-cited program, Social Stories (Gray 2004) was evaluated using a multi-element design with generalization probes across persons and setting.

Method

Participants and Setting Four participants (3 male, 1 female) between the ages of 5 and 9 years of age participated in the study. All participants were diagnosed with Autism Spectrum Disorder according to the DSM 5 (APA 2013) by at least two independent sources. The study was conducted at a behavior therapy clinic that each participant attended for a 2 h session once per week. The study was conducted in a playroom (2.6×4.4 m) that consisted of a small desk and chairs and a cabinet with toys and books. Sessions were videotaped to permit calculation of reliability measures.

Children were eligible to participate in the study if they had social deficits, as evidenced by direct observation and parent and therapist report, had no prior exposure to either intervention, and completed a prerequisite inventory, demonstrating the ability to a) focus on a story or video for 30 s, b) answer comprehension questions, and c) demonstrate non-vocal and vocal imitative repertoires, all according to direct assessment at the clinic.

At the beginning of the study, Sam was a 6-year, 7-month-old Caucasian boy with autism with a standard score of 79 on the Peabody Picture Vocabulary Test-III (PPVT-4; Dunn and Dunn 2007) and a standard score of 85 on the Vineland Adaptive Behavior Scale (Vineland; Sparrow, Balla, and Cicchetti 1984). Sam

rarely initiated social interaction with others and gave eye contact infrequently. He often played interactive games appropriately with prompting and exhibited some stereotypical behavior with toys that spin or made noise.

Max was an 8-year, 9-month old Caucasian boy with autism with a standard score of 87 on the PPVT and a standard score of 75 on the Vineland. Max exhibited fleeting eye contact, made limited and atypical social initiations, and did not engage in social interaction. His initiations to others were based on topics of perseverative interest.

Mike was a 6-year, 4-month-old Latino-Hispanic boy with autism with a standard score of 64 on the PPVT and a standard score of 91 on the Vineland. He displayed poor eye contact, engaged in echolalia, and consistently responded to others' communicative bids, but rarely initiated to others. Mike played with toys in stereotypical ways and with obsessive objects of interest.

Kendra was a 6-year, 10-month-old Filipino girl with autism with a standard score of 68 on the PPVT and a standard score of 78 on the Vineland. Kendra consistently responded to her name being called with eye contact. She also frequently initiated social interactions, though often paired with echolalia and on topics of perseverative interest.

Materials Materials included videos and Social Stories that were individualized for each participant and their target behaviors. Videos were created as outlined by Charlop and Milstein (1989). The adult models were two females and one male who interacted with the child's interactive partner in a brief video clip (20–45 s). These videos were played for each participant through Windows Media Player on a 15 in. laptop screen.

The Social Stories were derived from Gray and Arnold's (2000) sample stories and Gray's (2004) Social Story guidelines. Per the guidelines, each story contained at least one of each of the following sentences: a) descriptive, which describes the context, b) directive, which defines the desired behavior, c) perspective, which focuses on others' reactions and emotions, and d) affirmative, which focuses on the shared beliefs of a culture. The stories were printed in color on a 8 ½×11 sheet of paper and included one Clipart picture relevant to the story at the top of the page (e.g., a child passing a ball to another child for a sharing story). Large print sentences were below the picture (size 16–24 Times New Roman font to allow the story to fit on a single page). Each story was presented to the child on a clipboard. See Appendix B for a sample Social Story.

Response Measurement Two specific social behaviors were selected for each participant based on observation and assessment of the child's social deficits prior to baseline. The specific social behaviors were counterbalanced across treatment condition and child and are shown in Tables 1 and 2.

Design A multi-element design and concurrent multiple baseline design across four participants was used to compare the efficacy of the two interventions, video modeling and Social Stories, in teaching social behaviors to children with ASD (Barlow and Hayes 1979; Kazdin and Hartmann 1978; Sindelar, Rosenberg, and Wilson 1985).

Table 1 Target behaviors and stimuli across condition and participant

Child	Video Modeling	Stimuli	Social Stories	Stimuli
Sam	Turn-Taking	Don't Break the Ice®	Greetings	Blocks
Max	Social Commenting	Blocks	Sharing	Puzzibits®; magnets; ViewMaster®
Mike	Greetings	Toot & Otto game; Blocks	Sharing	Puzzibits®; magnets; ViewMaster®
Kendra	Sharing	Dora® See 'n Say; Doodle Pad®; Etch-a-Sketch®	Social Commenting	Toot & Otto game; Blocks

Procedure

Assessment Sessions An assessment session was a 5 min play session in which the target behavior was observed and scored; these sessions occurred in baseline and after each intervention session. The same conditions were used regardless of the phase of the study: during baseline, intervention, and follow-up.

Table 2 Experiment 1 – Operational Definitions of Target Behaviors

Target behavior	Operational definition
Sharing	Sharing consists of: 1. Vocal inquiry (e.g., “Do you want a turn?”) <i>(Partner says yes)</i> 2. Handing the item to the individual within 10 s
Social commenting	Social Commenting • Responding to individual being hurt: (e.g., “Oh no, are you okay?”) • Responding to individual losing (e.g., That’s okay. Let’s play again.”) • Responding to individual knocking over toy and showing disappointment (e.g., “Oh no, I’m sorry that happened.”)
Greetings	Greetings consist of: 1. Looking toward an individual entering the room 2. Responding within 10 s of the individual entering the room (e.g., “hello”, “hi”, or “hi there”)
Turn-taking	Turn-taking consists of: 1. Relinquishing any necessary items when appropriate 2. Waiting during another person’s turn 3. Identifying one’s own turn within 10 s of another’s turn (e.g., “my turn”)

Note: Each of the numbered behaviors was required for a correct response; however, only one of the social commenting behaviors (bullets) was required, depending on the bid given

During each 5 min assessment session, an interactive partner provided a social bid one time per minute. These sessions were conducted immediately following intervention sessions (either video modeling or Social Stories). A familiar adult served as the interactive partner. If participants responded independently within 20 s of the experimenter's bid, the response was scored correct; otherwise, an incorrect response was scored. The criterion for mastery was two consecutive sessions in which the behavior occurred with a minimum of 80 % average.

During assessment sessions, the interactive partner provided an opportunity for the child to engage in their target behavior once per minute. For example, for a target behavior of sharing, the partner expressed interest in the activity in which the child was engaged by saying, "that toy looks like so much fun." If the child engaged in the target behavior within 20 s of the experimenter's bid, the behavior was scored as correct. If an inappropriate response occurred, it was not scored and the trial continued until the interval ended. After correct responses, the partner provided a vocal response (e.g., said "thanks" if the child offered a toy to share, or stated "I'm okay, thanks"), providing no additional feedback regarding the child's behavior. This response aimed to mimic the contingencies in the natural environment and was provided when a child exhibited the target behavior across all conditions (including during baseline and maintenance probes). Any target behavior that occurred 20 s or more after the experimenter's bid was considered a spontaneous social behavior and was not scored as correct in response to the opportunity; the interactive partner responded similarly for children's spontaneous social behaviors.

Video Modeling Participants watched a short video of models demonstrating the individualized target behavior. During each intervention session, the participant sat at a small table and watched the video on a laptop screen approximately 1–2 ft away from them. Before the video began, the experimenter instructed the child to "watch the movie." During the video, if the child did not attend to the video for two consecutive seconds, the experimenter paused the video and redirected the child to attend. The experimenter played the video two times. Immediately after watching the video twice, an assessment session started, and the participant was given the opportunity to respond to social bids to see if the learning criterion was met. If not, video modeling intervention continued; if the learning criterion of two consecutive sessions (across 2 days) in which the behavior occurred with a minimum of 80 % average was met, intervention was discontinued. This process of presenting intervention, then assessing for criterion continued until criterion was met. No more than two intervention sessions were conducted in one day.

Social Stories Participants listened to a brief story about the individual target behavior and social situation. Children sat at a small table and the experimenter stated, "Let's read a story about [title of story]." The experimenter read the story that described the participant's appropriate individualized target behavior. If the child did not attend to the story, the experimenter paused and redirected the child to attend. After reading the story, the experimenter asked one simple comprehension question to assess their attention to the story (e.g., "what should you do with your friends?" A response of "share with them" was considered correct.) If a participant answered a comprehension question incorrectly, the experimenter repeated the question and prompted a correct

response. The experimenter then represented the question, and the session ended after the correct response; participants made no more than 1 error on comprehension questions per session, and a correct, independent response was required to proceed to an assessment session. During each intervention session, the social story was read twice; three of the four participants listened to the story both times, and one participant read the story the second time. Immediately after being presented with the story twice, an assessment session started, and the participant was given the opportunity to respond to social bids to see if the criterion was met. No more than two intervention sessions were conducted in one day.

Generalization and Maintenance Probes to assess generalization across unfamiliar people and novel settings were conducted both during baseline and after learning criterion was met. Generalization probes were taken in novel settings (e.g., café or courtyard) and with unfamiliar people. Maintenance of skills was assessed between one and 2 months after the completion of intervention. Intervention materials (video or story) were not present during generalization or maintenance probes, and participants had no exposure to the story, video, or target situations after the conclusion of treatment.

Inter-Rater Reliability Two research assistants observed video-recorded sessions to gather reliability data. They were trained to 90 % agreement on target behavior definitions and the procedural fidelity checklist with an experimenter and one other research assistant. Inter-observer reliability was calculated for 40 % of all sessions by dividing the number of agreements by the number of agreements plus disagreements multiplied by 100. IOA was an average of 95 % across participants and conditions (range=89–100 %). To adjust for chance agreement for reliability data, Cohen's kappa was calculated and $k=0.90$ (Cicchetti 1994; Cohen 1960).

Procedural Fidelity Research assistants calculated procedural fidelity on 40 % of sessions by scoring experimenter adherence to an 18-item checklist; the total number of intervention steps implemented correctly was divided by the total number. Procedural fidelity for individual sessions averaged 90 % (range: 81–100 %).

Results and Discussion

Results are shown in Fig. 1. For all four children, the social behaviors were at or near zero percentage during baseline. Upon introduction of the multi element comparisons, all four children met criterion and learned their social behaviors during the video modeling conditions, but failed to learn their social behaviors when presented with Social Stories. For two of the children, Mike and Kendra, video modeling was presented for the two social behaviors attempted in their Social Stories treatment condition. As can be seen in Fig. 1, both children reached criterion with video modeling for these two previously unlearned behaviors. Additionally, there are some positive findings of generalization to new settings and unfamiliar people.

In summary, these results replicate previous results on the efficacy of video modeling and expand that literature to additional social behaviors. As well, this study

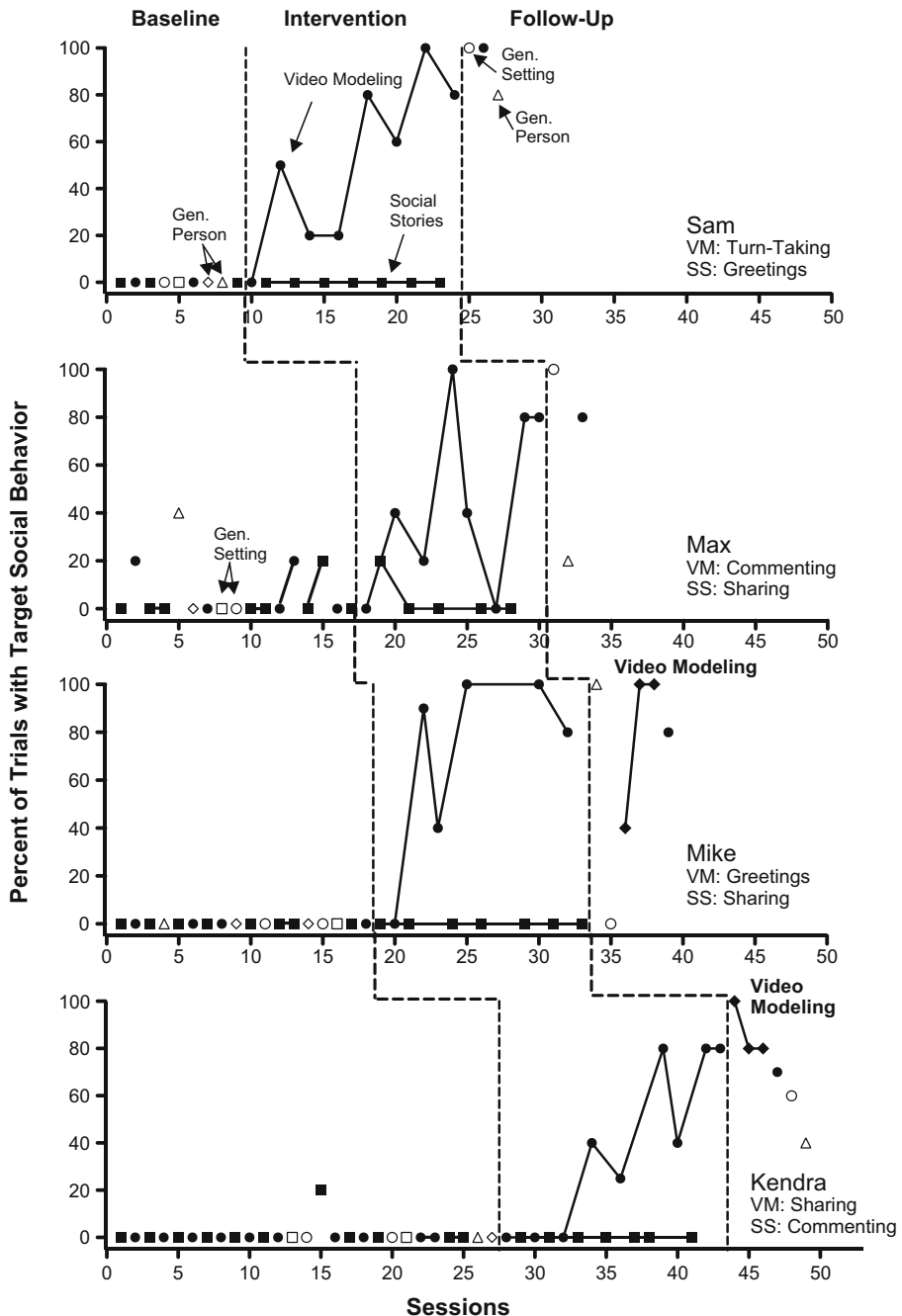


Fig. 1 Percentage of opportunities of target behaviors in video modeling and Social Stories conditions for all four participants in Experiment 1

provides a direct treatment comparison of video modeling with another intervention and demonstrates not only that video modeling is the more effective procedure with

these participants with ASD, but that there are a lack of treatment gains with Social Stories.

These results might initially seem surprising when compared with previous literature on Social Stories (e.g., Crozier and Tincani 2005; Dodd et al. 2008; Sansosti and Powell-Smith 2006; Scattone et al. 2006). However, recent information has suggested that Social Stories is not as effective as popular press would dictate (Kokina and Kern 2010; Reynhout and Carter 2006; Test et al. 2011). Indeed, recent reviews and meta-analyses have shown mixed findings of the efficacy and low clinical significance of Social Stories. Authors of meta-analyses have pointed out the considerable variability in outcomes for individuals. Of the 47 participants included in Kokina and Kern's (2010) meta-analysis, Social Stories was categorized as *highly effective or effective* for 24 of the individual outcomes and *ineffective* for 21 of the individual outcomes. In their analysis of PND data from the Social Stories literature, Reynhout and Carter (2006) did not find consistent effects of behavior change for individuals with ASD.

It appears that much of the research on Social Stories has used treatment packages and included behavioral components in combination with the social story rendering it difficult to determine the individual contribution of singular variables (e.g., Barry and Burlew 2004; Chan and O'Reilly 2008; Crozier and Tincani 2007; Thiemann and Goldstein 2001). Experiment 2 will address this issue.

Experiment 2

While Social Stories enjoy much popularity (e.g., Gray and Garand 1993; Gray 2004), upon close inspection of the literature, it appears that the positive findings are not as robust as initially thought (Reynhout and Carter 2011b; Styles 2011; Test et al. 2011). In a recent meta-analysis, Kokina and Kern (2010) suggested that other mechanisms may actually be at work, Chan and O'Reilly (2008) reported a large variation in the manner in which "social stories" are actually presented. Additionally, Social Stories have been found to be more effective in targeting behavior reduction than acquisition of social skills, and the small clinical significance of this behavior change has recently been questioned (Kokina & Kern; Reynhout and Carter 2006; Reynhout and Carter 2011a). The second experiment was designed to provide another treatment comparison – this time, comparing Social Stories, in isolation and in the form closely resembling the original (Gray and Garand 1993; Gray 2004), with behavior procedures often included in Social Stories treatment packages.

Method

Participants and Setting Participants were two children with a diagnosis of Autism Spectrum Disorder from two independent sources (DSM 5, American Psychiatric Association 2013). Participants were recruited with flyers from the community; inclusion criteria were that the child was between the ages of 4 and 10 years old, had an existing verbal repertoire, and demonstrated significant social skill deficits. Each participant's target behaviors were selected based on direct assessment of social skills. Research sessions were conducted during 2 hour weekly visits to a university-based research clinic. All sessions were conducted in 8' × 10' room

containing a table and two small chairs and that was equipped with video cameras and a two-way mirror.

Ryan was a 6-year, 5-month old boy who had an extensive vocal repertoire and displayed immediate and delayed echolalia. His PPVT-4 (Dunn and Dunn 2007) standard score was 81, and his Vineland Adaptive Behavior Scale (Vineland-II; Sparrow et al. 2005) scores were 88 (Communication domain) and 58 (Socialization domain). Antonio was an 8-year, 11-month old boy who spoke in full sentences and engaged in high levels of vocal stereotypy. His PPVT-4 standard score was 75, and his Vineland scores were 91 (Communication) and 56 (Socialization).

Materials Social Stories were written according to Gray and Garand's (1993) guidelines (further described by Ali and Frederickson 2006 and Gray 2004). An independent researcher analyzed the content of each Social Story to confirm that all Social Stories fit the basic Social Story ratio and were composed of the following types of sentences: descriptive sentences (expressing a fact), perspective sentences (describing an individual's thoughts or feelings), directive sentences (describing desired responses to social situations), and affirmative sentences (expressing a commonly shared value/opinion). The text was presented (font size 22) on an $8^{1/2} \times 11$ in. piece of paper with one Clipart picture that corresponded to the target behavior (e.g., a boy passing a toy to a girl for the sharing story). Stimuli used for Ryan's target behavior of helping included a snack container, a mechanical pencil, a wipes container, a computer power cord; four generalization probe stimuli were a water bottle, a watch, a sweat shirt, and a clip. The stimulus set used for Antonio's target behavior of empathetic statements was a Chutes and Ladders™ board game; the stimulus set for generalization probes was the Hi Ho! Cherry-O™ board game

Response Measurement The dependent variables were specific social behaviors; see Table 3 for operational definitions. The percentage of correct responses was calculated by dividing the number of correct responses by the number of opportunities to respond and multiplying by 100.

Table 3 Experiment 2 – Operational Definitions of Target Behaviors

Participant	Target behavior	Operational definition
Ryan	Offering to Help	A contextually appropriate verbal and/or motor response providing assistance to another person.
Ryan	Reciprocal Commenting	A contextually appropriate statement or question (Wh-question) following a statement made by another person.
Antonio	Empathic/ congratulatory Statements	A contextually appropriate verbal response to affect expressed by another person.
Antonio	Reciprocal Questions	A contextually appropriate answer and a Wh-question following a question asked by another person.

Design A multiple baseline across participants and within participants across behaviors design was used to examine the effectiveness of Social Stories in isolation with prompting and reinforcement on social skills acquisition. Two social behaviors were selected for each participant; target behaviors were randomly assigned to their first treatment. If the target behavior did not show evidence of acquisition (i.e., greater than 25 % increase from baseline average) within ten sessions in the first intervention condition, the second condition was implemented.

Procedure

Baseline During baseline, each participant was presented with 5 opportunities to engage in the target behavior during a 10 min play session. If the child engaged in the target behavior (e.g., offered to help), the partner responded verbally with a natural consequence (e.g., “thanks, that’s nice of you”). No feedback was provided if the participant did not respond or responded incorrectly (similar to intervention phases).

Social Stories In this condition, the experimenter read the individualized story to the participant two times. The experimenter then asked two comprehension questions relevant to the target behavior in the story. Praise was delivered contingent upon a correct response, and the experimenter corrected the response if the participant responded incorrectly or did not respond within 5 s. Immediately after the comprehension questions, the participant was given an opportunity to respond to a discriminative stimulus. The partner responded verbally with a natural consequence contingent upon a correct response (similar to baseline). No feedback was provided if the participant did not respond or responded incorrectly.

Prompting A progressive time delay verbal prompting procedure was implemented. Following the delivery of the statement of interest (i.e., the discriminative stimulus), the experimenter provided a verbal prompt according to a prompt delay sequence (Charlop, Schreibman, and Thibodeau 1985; Halle, Marshall, and Spradlin 1979; Ingenmey and Van Houten 1991). Immediate prompts (0 s delay) were delivered for the first three sessions, then progressed to a 2 s delay; the prompt delay increased by 2 s increments, up to a maximum of 6 s. Feedback was not provided if the participant did not respond or responded incorrectly, and a natural consequence was provided contingent upon a correct response (similar to baseline).

Generalization and Maintenance Probes to assess generalization across unfamiliar people and novel settings were conducted both during baseline and after learning criterion was met. The experimental conditions that were in effect during baseline occurred during generalization probes. Generalization across setting was assessed in a large playroom. Between 1 and 2 months after the completion of intervention, maintenance of skills was assessed at follow-up. Intervention procedures were not used during generalization or maintenance probes, and participants had no exposure to the story or target situations after the conclusion of treatment.

Reliability Reliability data were collected for 40 % of all sessions by independent observers. Agreement coefficients were calculated by dividing the number of agreements by agreements plus disagreements and multiplying by 100. Mean inter-observer agreement for offering to help was 100 %, reciprocal commenting was 100 %, empathy was 97.8 % (range: 66 to 100 %), and reciprocal questions was 95 % (range: 80 to 100 %).

Results and Discussion

Results of Experiment 2 are shown in Figs. 2 and 3. The results indicated that for both participants, Ryan and Antonio, learning criterion was only achieved in the prompting conditions. Generalization for social skills learned via prompting, as well as follow-up, were also favorable. Follow up did seem to deteriorate for Antonio over time. No social behaviors met criterion with Social Stories and thus prompting needed to be implemented for those social skills that were initially assigned to the Social Stories condition for both boys.

The results of Experiment 2 are consistent with the literature suggesting mixed results on the efficacy of Social Stories (e.g., Kokina and Kern 2010; Scattone et al. 2006; Wang and Spillane 2009). Additionally, some literature has suggested that Social Stories procedures have actually been found to be more effective in targeting behavior reduction than acquisition of social skills (Kokina & Kern; Reynhout and Carter 2006; Reynhout and Carter 2011a).

General Discussion

In the present study, two experiments provided treatment comparisons while teaching social skills to children with ASD. In Experiment 1, video modeling was compared in a multi-element design with Social Stories and found to be effective and the superior intervention. In the treatment comparison in Experiment 2, prompting, via a delayed cue procedure, yielded positive results whereas Social Stories did not. Thus, the present study continues to provide and extend an evidence base for video modeling when teaching social skills to children with ASD. Further, a delayed cue prompting procedure is also functional to occasion social skill behavior change in the present study. However, the Social Stories program was not effective in both experiments.

It is important to be cautious when interpreting results. This study adds to a body of literature of mixed results on the effectiveness of Social Stories (Iskander and Rosales 2013; Kokina and Kern 2010; Test et al. 2011; Reynhout & Carter 2006; Reynhout and Carter 2011a; Scattone et al. 2006; Wang and Spillane 2009). Specifically, Social Stories, in isolation, was not effective, and so the present results suggests that research needs to tease out when Social Stories is a separate treatment or when it is used as part of a treatment package. What is especially needed is a pure component analysis of social stories treatment packages to determine the exact functional variables responsible for behavior change. More information as to whether or not behavior change is due to the story, or prompting, or reinforcement is important to

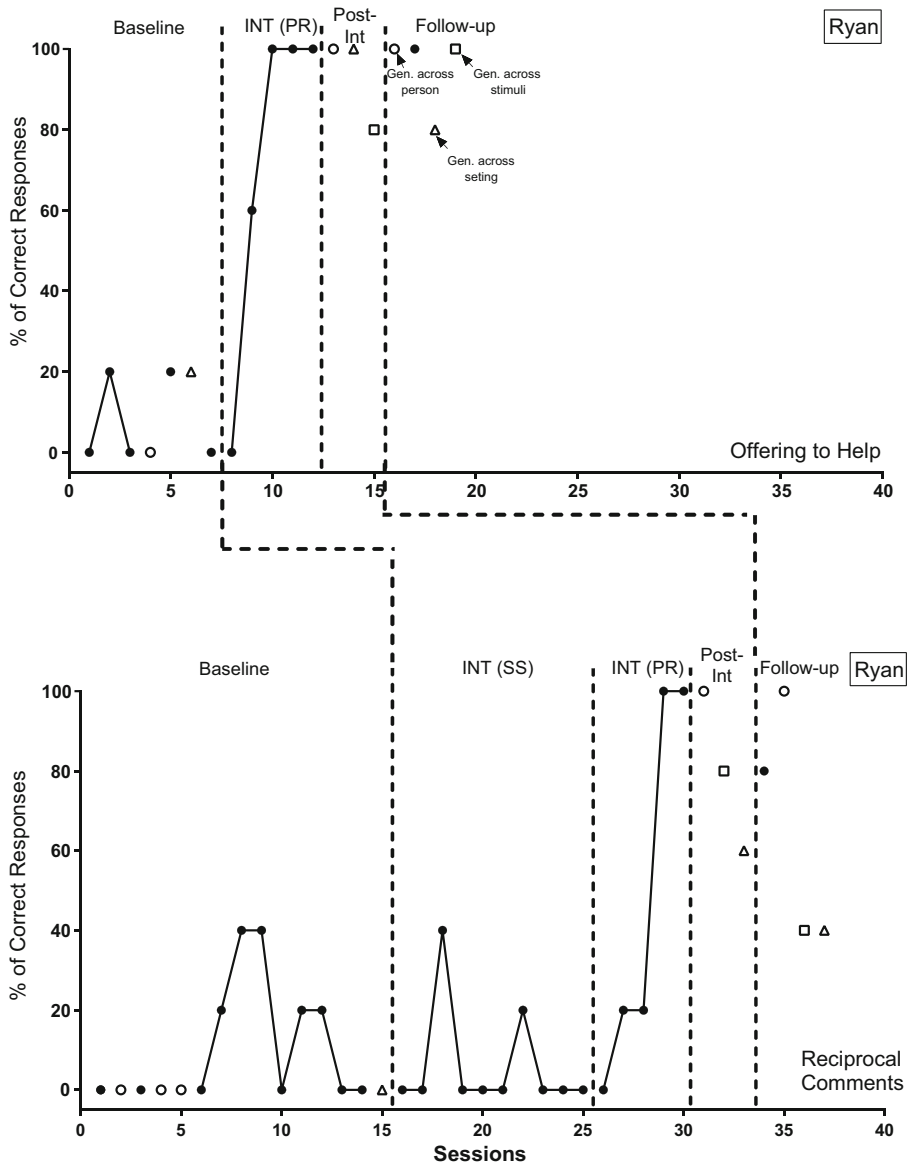


Fig. 2 Percentage of correct responses during Social Stories, prompting, and follow up conditions for Ryan in Experiment 2

know. For example, there is already a large body of evidence showing that imitation, prompting, priming and positive reinforcement are effective methods of teaching children with ASD (Cooper, Heron, and Heward 2007). These procedures are often combined with a Social Story intervention and so it is difficult to identify if the story alone is working or whether it is the combination of tactics (Kokina and Kern 2010; Reynhout and Carter 2009; Sansosti, Powell-Smith, and Kincaid 2004).

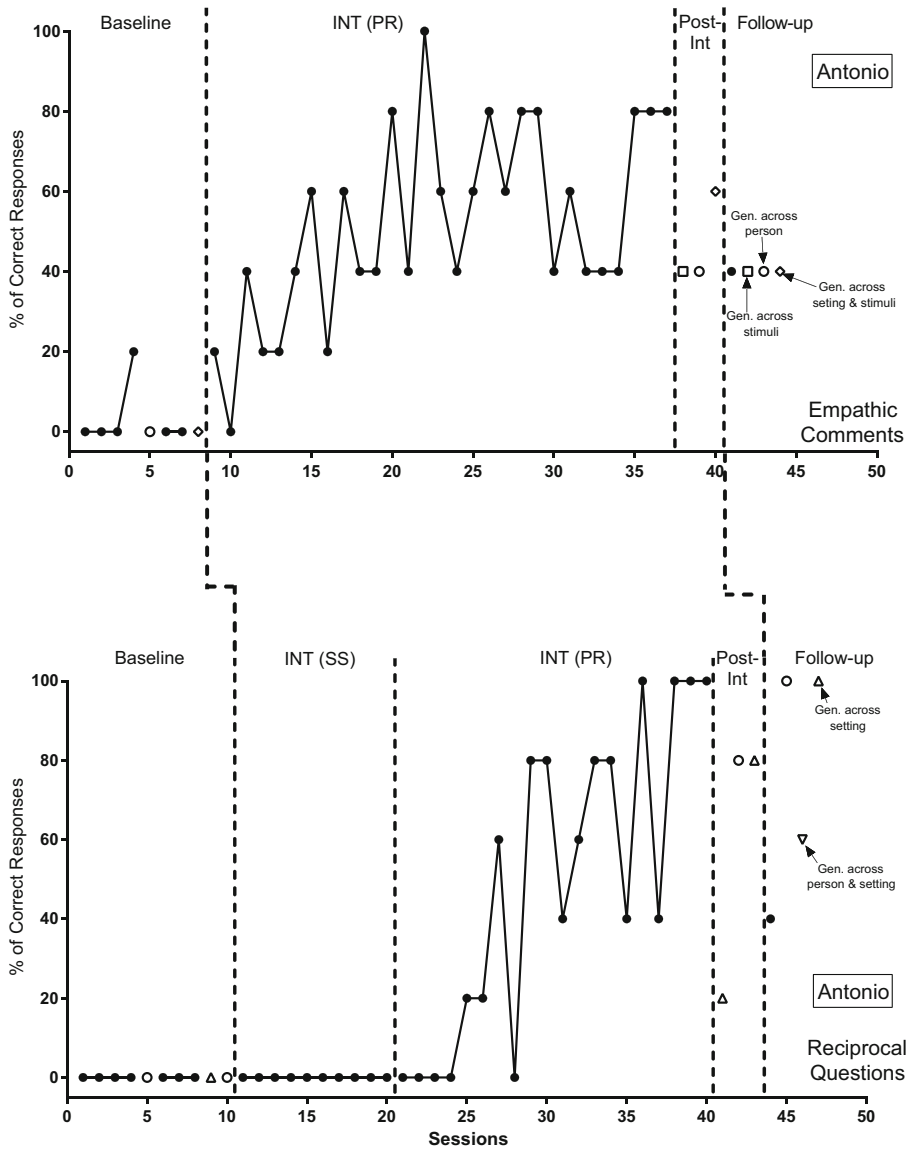


Fig. 3 Percentage of correct responses during Social Stories, prompting, and follow up conditions for Antonio in Experiment 2

Along this vein, the age range, verbal ability, and functioning level of the child participating may be important variables to consider when using Social Stories alone or in a treatment package. Kokina and Kern (2010) suggest that if a child has difficulty reading or understanding spoken language, it is much less likely that Social Stories will be effective than if the child has strong language abilities and can read. This may explain why there are mixed results in the Social Stories literature (Chan and O'Reilly 2008; Kokina and Kern 2010). Quirnbach, Lincoln, Feinberg-Gizzo, Ingersoll, and Andrews (2009) found that Social Stories for improving board game play in children

with ASD were only evident with those who were within or above the borderline range of the Verbal Comprehension Index (VCI) scores from the Wechsler Intelligence Scale for Children-Third Edition (WISC-III).

Furthermore, authors of recent meta-analyses investigating the use of Social Stories with individuals with ASD have found considerable variation in the effectiveness of the intervention and have concluded that Social Stories appear to produce only a small effect on behavior (Kokina and Kern 2010; Reynhout and Carter 2009, 2011a, b; Test et al. 2011). In a their meta-analysis of 62 Social Story studies, Reynhout and Carter (2009) concluded their paper stating that ‘while there was considerable variation, on average, Social Stories appear to have only a small clinical effect on behaviour and practitioners should factor this consideration into decisions about appropriate interventions’ (p. 13). Because of findings such as these, Test et al. (2011) concluded that Social Stories are not yet an evidence-based practice.

Both treatments compared to Social Stories in the present study, video modeling and delay cue prompting, have significant evidence bases for use with children with ASD. The efficacy of delayed cue prompting has been entrenched in our behavioral literature (e.g., Charlop et al. 1985; Cooper, Heron, and Heward 2007; Halle et al. 1979; Ingenmey and Van Houten 1991). Delay cue prompting has been used in conjunction with Social Stories, as a social stories treatment package. Video modeling has been hypothesized to be effective due to the salient visual cues provided (Charlop-Christy and Jones 2006; Charlop-Christy et al. 2000; Garretson, Fein, and Waterhouse 1990; Nikopoulos 2007; Rayner, Denholm, and Sigafoos 2009). The videos are designed to provide clear models of appropriate behavior that are easily repeated and are consistent across multiple presentations (Charlop-Christy and Jones 2006; Schreibman 2005). Reynhout and Carter (2011b) stated that, “one possible advantage of the use of video over text is that many social cues may be seen in relatively realistic context, as opposed to being described” (p. 374). With this in mind, it makes sense that VM is the superior treatment over Social Stories.

There are several limitations of the present investigation. In Experiment 1, in order to do the multi-element comparisons, a different social skill was used for each treatment condition. Whereas these social skills were randomly assigned, and the same social skill was used for both treatment conditions (i.e., sharing was assigned to both video modeling and Social Stories conditions), there was not an even counterbalance of social skills across treatment conditions. Also in Experiment 1, two of the participants who did not acquire their social skills via Social Stories were subsequently provided with a video modeling treatment to learn the social skill, a return to baseline reversal condition would have been a prudent addition to the design of the study. However, it is important to point out that since no learning took place during Social Stories, no return to baseline would have occurred. In Experiment 2, there are order effects in that prompting was effective with or without Social Stories occurring before its presentation. However, there is not a condition of prompting first, Social Stories second (because prompting was always effective when presented first). For both experiments, it is recommended that replication is always important for generalization of results across more participants.

As described in Experiment 1, video modeling procedures are presented to children who have a history of watching TV, videos, and/or watching movies on a tablet device. Also, attending to the screen is an important prerequisite skill. It

is important to note that Social Stories may also have important prerequisites that are not generally discussed in the literature. It is clear that for use of Social Stories, children with ASD should have some skills that should be considered before using this more abstract and complex form of treatment (Kokina and Kern 2010). Finally, more treatment comparison studies are called for in the literature so that researchers, teachers, and practitioners alike can see how various interventions weigh up against each other, in the quest for evidence-based treatments for children with ASD.

Acknowledgments This research was supported by the College of Social and Behavioral Sciences, California State University, Northridge. The authors would like to thank Amanda Finch, James Vogel, Rachel Travolta, Gina Chang, Kari Berquist, Sabrina Daneshvar, Alissa Greenberg, Denise Grosberg, Julyana Garcia, the research assistants at the CSUN Autism Clinic, and the children and families who participated.

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